

Workshop

The musical brain: Beat, rhythm, and timing



Tuesday June 7, 14.00-17.00 h
Doelenzaal (C0.07), Singel 425, Amsterdam

When we hear music, we often tap our feet, dance together or play and sing together. In order to do this, we need to be able to synchronize our behavior to the regular beat that is often present in music. This seems an easy task, but how do our brains accomplish this? Can everyone do this? And is this specific to humans?

Program

14.00	**Jessica A. Grahn (Western University, Canada)** *Music, movement, and the brain*
14.30	**Sonja A. Kotz (Maastricht University & Max Planck Institute, Leipzig, Germany)** *Timing and synchronization in sensorimotor networks of the brain*
15.00	**Rebecca S. Schaefer (Leiden University)** *What to do with that beat you found: cognitive aspects of moving to rhythm*
15.30	Coffee break
16.00	**István Winkler (Hungarian Academy of Sciences, Hungary)** *Auditory processing capabilities supporting communication in preverbal infants*
16.30	**Henkjan Honing (University of Amsterdam)** *Rhythm perception in birds and monkeys: some preliminary results*
17.00	Drinks

The workshop is organized on the occasion of the public PhD defense of Fleur Bouwer on Wednesday, June 8, at 11.00 in the Aula of the University of Amsterdam. You are cordially invited to attend the defense ceremony. For more information, visit www.fleurbouwer.nl.



INSTITUTE FOR LOGIC, LANGUAGE AND COMPUTATION

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Abstracts

Music, movement, and the brain

Jessica A. Grahn (Brain and Mind Institute & Dept. of Psychology, Western University, London, ON, Canada)

Moving to musical rhythm is an instinctive, often involuntary activity, but how does the brain produce this behaviour? In this talk I will describe how perception of musical rhythms activates motor brain areas even when no overt movement is made. I will cover studies of differences in rhythmic ability across people, examining how motor and auditory activity in the brain relate to how well different individuals perceive rhythm.

Timing and synchronization in sensorimotor networks of the brain

Sonja A. Kotz (Faculty of Psychology and Neuroscience, Dept. of Neuropsychology and Psychopharmacology, Maastricht University; Dept. of Neuropsychology, Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany)

Temporal prediction in action, perception, and cognition is a prerequisite for the effective interaction of an individual and the environment. It relies on an internal representation of temporal structure, likely involving the cerebellar and basal ganglia temporal processing systems of the brain (see Buhusi and Meck, 2005). In turn, a representation of temporal structure can be considered instrumental in synchronizing not only motor, but also sensory, and cognitive processes, and the allocation of cognitive recourses to external events. In my talk I will present behavioral, EEG, and fMRI data in support of these ideas by looking into sensorimotor synchronization and temporal adaptation, temporal prediction, subjective rhythmitization, and rhythm perception in different modalities in healthy and patient populations.

What to do with that beat you found: cognitive aspects of moving to rhythm

Rebecca S. Schaefer (Health, Medical and Neuropsychology Unit, Institute of Psychology, Leiden University)

Although synchronisation of movement to an auditory rhythm is often described as an easy, near-automatic task, spontaneous synchronisation is usually not very precise, and the skill of being 'musically in time' is generally perfected over years of training by expert musicians. Moreover, there are strong indications that moving to rhythm requires attentional resources, which are more readily available to healthy people than to patient groups. As rhythmic auditory cues are increasingly used in rehabilitation settings, a better understanding of the cognitive demands of moving to sound can inform better intervention design. I will give a brief overview of some recent work focusing on this issue, with consideration for both the practical and theoretical aspects of moving to rhythm.

Auditory processing capabilities supporting communication in preverbal infants

István Winkler (Research Centre for Natural Sciences, Hungarian Academy of Sciences)

It is well-known that infants can enter dialogues using sounds well before they learn to speak and that communication with adults is essential for socio-cognitive development. The theory of natural pedagogy postulates the existence of a human-specific form of teaching in which adults convey to infants precise information about objects in the environment and solutions to various problems (ostensive communication). This type of information transfer contrasts the goal-oriented learning observed in animals (with the possible exception of dogs, but not non-human primates). Ostensive communication requires infants to carry dialogues with adults, detecting when being addressed, marking understanding, consent, etc. In the talk, I suggest that the advanced auditory processing capabilities observed in preverbal infants are precisely the ones needed to support dialoguing by allowing infants to a) segregate concurrently active sound sources, b) represent speaker identity, c) categorize messages (speech vs. non-speech), d) detect whether or not a message has been directed to them, e) determine whether some action or answer is required, and f) reply or interject with correct timing.

Rhythm perception in monkeys: some (very) preliminary results

Henkjan Honing (Institute for Logic, Language and Computation, Amsterdam Brain and Cognition, University of Amsterdam)

Beat-based timing in humans is a complex neurocognitive phenomenon that depends on a dynamic interaction between auditory and motor systems in the brain (Grahn & Brett, 2009; Merchant, Grahn, Trainor, Rohrmeier, & Fitch, 2015). It is hypothesized to be facilitated by bidirectional (and potentially causal) links between the auditory and motor areas in the brain, including the motor cortico-basal-ganglia-thalamo-cortical (mCBGT) circuit, that appear to be more developed in humans as opposed to nonhuman primates and related species (Merchant et al., 2015).

These observations lead to the *gradual audiomotor evolution hypothesis* (GAE; Merchant & Honing, 2014) that suggests beat-based timing to be gradually developed in primates, peaking in humans but present only with limited properties in other nonhuman primates, while humans share interval-based timing with all nonhuman primates and related species. Thus, this hypothesis accommodates the fact that the performance of rhesus monkeys is comparable to humans in single interval tasks (such as interval reproduction, categorization, and interception), but differs substantively in multiple interval tasks (such as rhythmic entrainment, synchronization and continuation).

We will present some preliminary results of a rhythm perception study in two rhesus monkeys (*Macaca mulatta*) in support of the GAE hypothesis, by using a novel paradigm (Bouwer, Werner, Knetemann, & Honing, 2016) that allows for testing the various contributions of regularity and beat perception on the temporal predictability of a rhythmic stimulus (Honing, Bouwer, & Háden, 2014).